Anthony Meza

Ph.D. Candidate

Woods Hole, MA • (714) 552-2396 • ameza@mit.edu • anthony-meza.github.io

EDUCATION

Massachusetts Institute of Technology & Cambridge, MA
Woods Hole Oceanographic Institution *Expected Nov 2026*
Ph.D. in Physical Oceanography | Advised by Dr. Geoffrey Gebbie
Dissertation: "Forced Responses of the Global Ocean Circulation and Implications for Tracer Distributions"

University of California, Irvine Irvine, CA
B.S. in Mathematics *2018–2021*

PUBLICATIONS

- **Meza, A.**, & Gebbie, G. (2025). Wind-driven mid-depth Pacific cooling in a dynamically consistent ocean state estimate. *Journal of Geophysical Research: Oceans*. doi.org/10.1029/2025JC022462
- **Meza, A.**, Drake, H. & Gebbie, G. *(In prep)*. Sea ice-driven decline of abyssal ocean ventilation in a high-resolution coupled climate model.
- **Meza, A.**, Bhuyan, P., Zheng, Z. & Gebbie, G. *(In prep)*. Passive and dynamic contributions to mean ocean temperature during the Last Glacial Maximum.

RESEARCH EXPERIENCE

Woods Hole Oceanographic Institution Sep 2021–Present
Graduate Research Assistant *Woods Hole, MA*

- Designed and ran global ocean simulations using MITgcm to test mechanisms controlling deep ocean circulation and heat content
- Evaluated high-resolution coupled climate models to quantify the impacts of Antarctic sea ice melt on deep water formation and interior tracer distributions
- Used ocean reanalysis and satellite products to establish a statistical relationship between remotely-generated near-shore sea surface temperature variability and extreme precipitation events in California
- Developed Python and Julia tools to process, visualize, and interpret terrabytes of climate data in high-performance computing environments

Foundation for Resilient Societies Jan 2025
Technical Consultant *Cambridge, MA*

- Ran and debugged Strategic Energy & Risk Valuation Model (SERVM) simulations to assess adequacy of U.S. electrical grid capacity under varying generation scenarios (e.g., solar adoption).
- Led a team of 12 undergraduate electric grid modeling interns to develop an internal user guide for running SERVM experiments and interpreting model output.

Los Alamos National Laboratory Jun 2021–Aug 2021
Research Intern *Los Alamos, NM*

- Implemented and evaluated reduced-precision in the Energy Exascale Earth System Model (E3SM) to reduce computational cost and energy consumption in global climate simulations

Institute for Pure and Applied Mathematics & Jun 2020–Sep 2020
The Aerospace Corporation
Research Intern *Los Angeles, CA*

- Designed and implemented reinforcement learning–based methods for adaptive packet routing in satellite network simulations, implemented in Python using PyTorch

TECHNICAL PROJECTS

Atmospheric Physics using AI Apr 2024–Jul 2024
Kaggle Competition Participant *Remote*

- Ranked 70th percentile on the public leaderboard for emulating subgrid-scale atmospheric processes
- Developed machine learning architectures with physical constraints in PyTorch and TensorFlow
- Optimized data pipelines for GPU-accelerated training, significantly increasing model throughput

TEACHING EXPERIENCE

- Co-organizer and Instructor *2024*
  High Performance Computing and Data Analysis Workshop
- Calculus Instructor *2024*
  MIT-WHOI Joint Program Summer Math Refresher
- Partial Differential Equations Instructor *2023*
  MIT-WHOI Joint Program Summer Math Refresher

PROFESSIONAL ACTIVITIES

- Committee Member *2024–Present*
  AMS Committee on Climate Variability and Change
- Graduate Application Mentor *2023–Present*
  Joint Program Applicant Support & Knowledgebase
- Physical Oceanography Representative *2023–2024*
  WHOI Joint Program Student Representative
- Conference Co-Organizer *2023*
  2023 Graduate Climate Conference
- At-Large Representative *2022–2023*
  WHOI Joint Program Student Representative
- Conference Co-Organizer *2022*
  2022 First Generation Summit
- Committee Member *2020–2021*
  UC Irvine Mathematics Inclusive Excellence Committee

WORKSHOPS AND SUMMER SCHOOLS ATTENDED

- CESM/MOM6 Regional Modeling Workshop *May 2025*
  NCAR Mesa Laboratory, Boulder, CO
- ECCO Summer School *May 2025*
  Asilomar Conference Center, Monterey, CA
- Tracer Mixing in Fluids Across Planetary Scales Summer School *Jul 2024*
  Brin Mathematics Research Center, College Park, MD

PRESENTATIONS

- **A. Meza**, H. Drake, G. Gebbie. "Projected changes in Southern Ocean surface water mass transformation in an eddy-resolving coupled climate model" Ocean Sciences Meeting, 22–27 February 2026, Glasgow, Scotland. *Poster*.
- **A. Meza**, G. Gebbie. "Wind-driven mid-depth Pacific cooling in a dynamically consistent ocean state estimate" ECCO Summer School, 19–30 May 2025, Asilomar Conference Center, Monterey, CA. *Poster*.

- **A. Meza**, P. Bhuyan, Z. Zheng, G. Gebbie., M. Linz, J. Wenegrat. "Surface to Bottom Connections in Earth's Ocean" Tracer Mixing in Fluids Across Planetary Scales Summer School, 8–19 Jul 2024, Brin Mathematics Research Center, College Park, MD. *Talk*.
- **A. Meza**, H. Seo. "Associations Between Coastally Trapped Waves and Wintertime Precipitation in California" Ocean Sciences Meeting, 18–23 Feb 2024, New Orleans, LA. *Poster*.
- **A. Meza**, H. Seo. "Associations Between Coastally Trapped Waves and Wintertime Precipitation in California" Graduate Climate Conference, 1–3 Nov 2023, Marine Biological Laboratory, Woods Hole, MA. *Poster*.
- **A. Meza**, G. Gebbie. "Drivers of subsurface Pacific cooling in ECCOv4r4" ECCO Annual Meeting 2023, 25 Jan 2023, University of Washington, Seattle, WA. *Virtual Talk*.
- **A. Meza**, G. Gebbie. "Drivers of mid-depth Pacific cooling trends in an ocean reanalysis" AGU Fall Meeting 2022, 2–4 Nov 2023, Chicago, IL. *Poster*.
- **A. Meza**, G. Gebbie. "Drivers of mid-depth Pacific cooling trends in an ocean reanalysis" Graduate Climate Conference, 31 Oct 2022, University of Washington, Seattle, WA. *Poster*.
- C. Tran, **A. Meza**, H.L. Tung, H. Liu. "A Reinforcement Learning Approach to Packet Routing on a Dynamic Network" Joint Mathematics Meeting, 6–9 Jan 2021, Virtual. *Virtual Talk*.

AWARDS AND HONORS RECEIVED

- GEM Fellowship *2021*
  National Consortium of Graduate Degrees for Minorities in Engineers, MIT
- Rose Hills Scholarship *2020*
  Rose Hills Foundation, UC Irvine
- Rose Hills Scholarship *2019*
  Rose Hills Foundation, UC Irvine
- Bellettini Scholarship *2019*
  Maria Rebecca and Maureen Bellettini Fund, UC Irvine
- Southern California Edison STEM Scholarship *2019*
  Southern California Edison, UC Irvine

SKILLS

Languages: *Programming*: Python, Julia, Fortran, MATLAB; *Human*: English, Spanish
Scientific Computing: NumPy, SciPy, xarray, Pandas, xgcm, Optimization.jl, JuMP.jl, scikit-learn, PyTorch, TensorFlow, keras
HPC & Dev Tools: Unix/Linux, OpenMPI, HPC job schedulers (e.g., Slurm), Dask, Git, GitHub

PROFESSIONAL REFERENCES

Geoffrey Gebbie (ggebbie@whoi.edu)

Henri Drake (hdrake@uci.edu)

Christopher Piecuch (cpiecuch@whoi.edu)